

Online Convex Optimization with Unbounded Memory

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Abstract

Online convex optimization (OCO) is a widely used framework in online learning. In each round, the learner chooses a decision in some convex set and an adversary chooses a convex loss function, and then the learner suffers the loss associated with their chosen decision. However, in many of the motivating applications the loss of the learner depends not only on the current decision but on the entire history of decisions until that point. The OCO framework and existing generalizations thereof fail to capture this. In this work we introduce a generalization of the OCO framework, “Online Convex Optimization with Unbounded Memory”, that captures long-term dependence on past decisions. We introduce the notion of p -effective memory capacity, H_p , that quantifies the maximum influence of past decisions on current losses. We prove a $O(\sqrt{H_1 T})$ policy regret bound and a stronger $O(\sqrt{H_p T})$ policy regret bound under mild additional assumptions. These bounds are optimal in terms of their dependence on the time horizon T . We show the broad applicability of our framework by using it to derive regret bounds, and to simplify existing regret bound derivations, for a variety of online learning problems including an online variant of performative prediction and online linear control.

1 Introduction

Online learning models sequential decision-making problems, where a learner must choose a sequence of decisions while interacting with the environment. We refer the reader to [Cesa-Bianchi and Lugosi \[2006\]](#), [Shalev-Shwartz \[2012\]](#) for more background. One of the most popular frameworks in online learning is online convex optimization (OCO) (see, for example, [Hazan \[2019\]](#)).

The OCO framework is as follows. In each round, the learner chooses a decision from some convex set and an adversary chooses a convex loss function, and then the learner suffers the loss associated with their current decision. The performance of an algorithm is measured by regret: the difference between the algorithm’s total loss and that of the best fixed decision.

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This framework has numerous applications, such as prediction from expert advice [Littlestone and Warmuth, 1989, 1994], portfolio selection [Cover, 1991], routing [Awerbuch and Kleinberg, 2008], etc. There has been a lot of work in developing a variety of algorithms and proving optimal regret bounds, and we refer the reader to Hazan [2019] for a survey. In many applications of the OCO framework the loss of the learner depends not only on the current decisions but on the entire history of decisions until that point. For example, consider the problem of online linear control [Agarwal et al., 2019b], where in each round the learner chooses a “control policy” (i.e., decision), suffers a loss that is a function of the action taken by the chosen policy and the current state of the system, and the system’s state evolves according to linear dynamics. Note that the current state depends on the entire history of actions and, therefore, the current loss depends not only on the current decision but the entire history of decisions. The OCO framework fails to capture such long-term dependence of the current loss on the past decisions, as do existing generalizations of OCO that allow the loss to depend on a *constant* number of past decisions [Anava et al., 2015]. Although a series of approximation arguments can be used to apply the finite memory generalization of OCO to the online linear control problem, there is no OCO framework that captures the complete long-term dependence of current losses on past decisions.

Contributions. In this paper we introduce a generalization of the OCO framework, “Online Convex Optimization with Unbounded Memory”, that allows the loss in the current round to depend on the entire history of decisions until that point. We introduce the notion of p -effective memory capacity, H_p , that quantifies the maximum influence of past decisions on present losses. We prove a $O(\sqrt{H_1 T})$ policy regret bound and a stronger $O(\sqrt{H_p T})$ policy regret bound under mild additional assumptions about the history (Theorem 3.1). Both bounds use the notion of policy regret with respect to constant action sequences, as defined in Dekel et al. [2012]. Our bounds are optimal in terms of their dependence on the time horizon T , and we show that our results match existing ones for the special case of OCO with finite memory (Section 4.1). Finally, we show the broad applicability of our framework to derive regret bounds, and to simplify existing regret bound derivations, for online learning problems including an online variant of performative prediction and online linear control (Section 4.3 and 4.4).

Related work. The most closely related work to ours is the OCO with finite memory framework [Anava et al., 2015]. They consider a generalization of the OCO framework that allows the current loss to depend on a *constant* number of past decisions. There have been a number of follow-up works that extend the framework in a variety of other ways, such as non-stationarity [Zhao et al., 2022], incorporating switching costs [Shi et al., 2020], etc. However, none of the existing works go beyond a constant memory length. In contrast, our framework allows the current loss to depend on an *unbounded* number of past decisions.

Reinforcement learning [Sutton and Barto, 2018] is another popular framework for sequential decision-making that considers very general state-action models of feedback and dynamics. In reinforcement learning one typically measures regret with respect to the best state-action policy from some policy class, rather than the best fixed decision as in online learning and OCO. In the special case of linear control, policies can be reformulated as decisions while preserving convexity; we discuss this application in Section 4. Considering the general framework is an active area of research. For example Bhatia and Sridharan [2020]

provide only non-constructive upper bounds on regret.

We defer discussion of related work for specific applications to Section 4.

2 Framework

Let T denote the time horizon. Let $\mathcal{X}$ be a closed and convex subset of a Hilbert space, $\mathcal{H}$ be a Banach space, $A : \mathcal{H} \rightarrow \mathcal{H}$ and $B : \mathcal{X} \rightarrow \mathcal{H}$ be linear operators, and $f_t : \mathcal{H} \rightarrow [0, 1]$ be loss functions chosen by an oblivious adversary. The game between the learner and the adversary proceeds as follows. Let the initial history be $h_0 = 0$. In each round $t \in [T]$, the learner chooses $x_t \in \mathcal{X}$, the history h_t is updated according to $h_t = Ah_{t-1} + Bx_t$, and the learner suffers loss $f_t(h_t)$.

Consider a strategy that chooses the same action in every round, i.e. $x_t = x$ for some $x \in \mathcal{X}$ and for all rounds t . The history after round t for such a strategy is described by $h_t = \sum_{k=0}^{t-1} A^k Bx$, which motivates the following definition.

Definition 2.1. Given a function $f_t : \mathcal{H} \rightarrow [0, 1]$, the function $\mathring{f}_t : \mathcal{X} \rightarrow [0, 1]$ is defined by $\mathring{f}_t(x) = f_t(\sum_{k=0}^{t-1} A^k Bx)$.

We define the policy regret of a learner as the difference between its total loss and the total loss of a strategy that plays the best fixed action in every round [Dekel et al., 2012].

Definition 2.2 (Policy Regret). The policy regret of an algorithm $\mathcal{A}$ is defined as

$$R_T(\mathcal{A}) = \sum_{t=1}^T f_t(h_t) - \min_{x \in \mathcal{X}} \sum_{t=1}^T \mathring{f}_t(x). \quad (1)$$

2.1 Notation

We use $\|\cdot\|_{\mathcal{X}}$ to denote the norm associated with the Hilbert space $\mathcal{X}$ and $\|\cdot\|_{\mathcal{H}}$ to denote the norm associated with the Banach space $\mathcal{H}$. The operator norm for a linear L operator from space $\mathcal{U} \rightarrow \mathcal{V}$ is defined as $\|L\|_{\mathcal{U} \rightarrow \mathcal{V}} = \max_{\|u\|_{\mathcal{U}} \leq 1} \|Lu\|_{\mathcal{V}}$. For convenience, sometimes we simply use $\|\cdot\|$ when the meaning is clear from the context. For a finite-dimensional matrix, we use $\|\cdot\|_F$, $\|\cdot\|_2$, and $\sigma_j(\cdot)$ to denote its Frobenius norm, operator norm, and j -th singular value respectively. When we say that $\mathcal{H}$ is a sequence space with the $\|\cdot\|_p$ norm for $p \geq 1$, we mean that $\|(y_0, y_1, \dots)\|_{\mathcal{H}} = (\sum_{k=0}^{\infty} \|y_k\|^p)^{\frac{1}{p}}$.

2.2 Assumptions

We make the following assumptions on f_t and $\mathring{f}_t$.

A1 The functions f_t are L -Lipschitz continuous, i.e.,

$$|f_t(h) - f_t(\tilde{h})| \leq L\|h - \tilde{h}\|_{\mathcal{H}}.$$

A2 The functions $\mathring{f}_t$ are differentiable. We use $\nabla \mathring{f}_t(x)$ to denote its gradient at x . Furthermore, the gradients are bounded, i.e., for all $x \in \mathcal{X}$,

$$\|\nabla \mathring{f}_t(x)\| \leq L.$$

A3 The functions $\mathring{f}_t$ are convex, i.e., for all $x, y \in \mathcal{X}$,

$$\mathring{f}(y) \geq \mathring{f}(x) + \langle \nabla \mathring{f}_t(x), y - x \rangle.$$

Assumptions **A1**, **A2**, and **A3** are standard in the optimization literature. Given these, we assume the following feedback model for the learner.

A4 There exists a gradient oracle such that at the end of each round t , the learner receives the gradient $\nabla \mathring{f}_t(x_t)$.

Assumption **A4** is a standard assumption in the literature on first-order optimization methods. Before presenting our final assumption, we introduce the notion of p -effective memory capacity that quantifies the maximum influence of past decisions on present losses.

Definition 2.3 (p -Effective Memory Capacity). Consider an online convex optimization with unbounded memory problem specified by the Hilbert space $\mathcal{X}$, Banach space $\mathcal{H}$, and linear operators $A : \mathcal{H} \rightarrow \mathcal{H}$ and $B : \mathcal{X} \rightarrow \mathcal{H}$. For $p \geq 1$, the p -effective memory capacity of this problem is defined as

$$H_p(\mathcal{X}, \mathcal{H}, A, B) = \left(\sum_{k=0}^{\infty} k^p \|A^k B\|^p \right)^{\frac{1}{p}}. \quad (2)$$

When the meaning is clear from the context we simply use H_p to denote the p -effective memory capacity. One way to understand this definition is the following. Consider two sequences of decisions whose elements differ by no more than k at time k : $\|x_k - \tilde{x}_k\| \leq k$. Then the histories generated by the two sequences have difference bounded as $\|h - \tilde{h}\| = \|\sum_k A^k B(x_k - \tilde{x}_k)\| \leq \sum_k k \|A^k B\| = H_1$. A similar bound holds with H_p instead when $\mathcal{H}$ is a sequence space with the $\|\cdot\|_p$ norm. Therefore, the p -effective memory capacity is an upper bound on the difference in histories for two sequences of decisions whose difference grows at most linearly in time.

A5 The 1-effective memory capacity is well-defined, i.e., $H_1 < \infty$.

When $\mathcal{H}$ is a sequence space with the $\|\cdot\|_p$ norm for $p \geq 1$, then note that $H_p \leq H_1 < \infty$ is well-defined under assumption **A5**. In this case we obtain stronger regret bounds that depend on H_p instead of H_1 .

3 Regret Analysis

In this section we present a regret analysis for OCO with unbounded memory. The approach consists of writing the regret as

$$\begin{aligned}
R_T(\mathcal{A}) &= \sum_{t=1}^T f_t(h_t) - \min_{x \in \mathcal{X}} \sum_{t=1}^T \mathring{f}_t(x) \\
&= \sum_{t=1}^T f_t(h_t) - \mathring{f}_t(x_t) + \sum_{t=1}^T \mathring{f}_t(x_t) - \min_{x \in \mathcal{X}} \sum_{t=1}^T \mathring{f}_t(x) \\
&\leq \underbrace{\sum_{t=1}^T f_t(h_t) - \mathring{f}_t(x_t)}_{(a)} + \underbrace{\max_{x \in \mathcal{X}} \left(\sum_{t=1}^T \langle \nabla \mathring{f}_t(x_t), x_t - x \rangle \right)}_{(b)},
\end{aligned}$$

where the last inequality follows from the convexity of $\mathring{f}_t$ (assumption **A3**). The second term (b) can be bounded by using the follow-the-regularized-leader (FTRL) algorithm (Lemmas 3.1 to 3.3), which we present in the context of our framework in Algorithm 1. The main novelty comes from bounding the first term (a) by measuring the difference between the histories h_t and $\sum_{k=0}^{t-1} A^k B x_t$ in terms of the p -effective memory capacity (Lemma 3.4).

Let $R : \mathcal{X} \rightarrow \mathbb{R}$ be an α -strongly-convex regularizer. Since we assume $\mathcal{X}$ is a Hilbert space, such an α -strongly-convex regularizer exists. The FTRL algorithm [Abernethy et al., 2008, Shalev-Shwartz and Singer, 2006, Hazan, 2019] chooses x_t as the minimizer of the past losses plus a regularizer. In our framework, this corresponds to

$$x_t \in \arg \min_{x \in \mathcal{X}} \sum_{s=1}^{t-1} \langle \nabla \mathring{f}_s(x_s), x \rangle + \frac{R(x)}{\eta}.$$

For notational convenience, let $g_0(x) = \frac{R(x)}{\eta}$, $g_t(x) = \langle \nabla \mathring{f}_t(x_t), x \rangle$ for $t \geq 1$, and $x^* \in \arg \min_{x \in \mathcal{X}} \sum_{t=1}^T g_t(x)$.

Algorithm 1: FTRL

Input : Time horizon T , step size η , α -strongly-convex regularizer $R : \mathcal{X} \rightarrow \mathbb{R}$.

1 Initialize history $h_0 = 0$.

2 **for** $t = 1, 2, \dots, T$ **do**

3 Learner chooses $x_t \in \arg \min_{x \in \mathcal{X}} \underbrace{\sum_{s=1}^{t-1} \langle \nabla \mathring{f}_s(x_s), x \rangle}_{=g_s(x)} + \underbrace{\frac{R(x)}{\eta}}_{=g_0(x)}.$

4 Set $h_t = A h_{t-1} + B x_t$.

5 Learner suffers loss $f_t(h_t)$ and observes $\underbrace{\nabla \mathring{f}_t(x_t)}_{=\nabla g_t(x_t)}.$

6 **end**

Theorem 3.1. Consider an online convex optimization with unbounded memory problem specified by the decision space $\mathcal{X}$, history space $\mathcal{H}$, and linear operators $A : \mathcal{H} \rightarrow \mathcal{H}$ and $B : \mathcal{X} \rightarrow \mathcal{H}$. Let the regularizer $R : \mathcal{X} \rightarrow \mathbb{R}$ be α -strongly-convex and satisfy $|R(x) - R(x')| \leq D$ for all $x, x' \in \mathcal{X}$. Algorithm 1 with step-size η satisfies the following regret bound

$$R_T(\text{FTRL}) \leq \frac{D}{\eta} + \eta \frac{TL^2}{\alpha} + \eta \frac{TL^2}{\alpha} H_1. \quad (3)$$

If we set $\eta = \sqrt{\frac{D\alpha}{TL^2(1+H_1)}}$, then the regret bound satisfies

$$R_T(\text{FTRL}) \leq 3\sqrt{\frac{DL^2(1+H_1)}{\alpha}}\sqrt{T}. \quad (4)$$

When $\mathcal{H}$ is a sequence space with the $\|\cdot\|_p$ norm for $p \geq 1$, then the above bounds hold with H_p instead of H_1 .

We prove this theorem using a sequence of lemmas that formalize the approach mentioned at the start of this section. The proofs of Lemmas 3.1 to 3.3 are standard in the literature on FTRL, but we include them in the supplementary material for completeness (Appendix A).

Lemma 3.1. $\forall x \in \mathcal{X}$, $\sum_{t=0}^T g_t(x) \geq \sum_{t=0}^T g_t(x_{t+1})$.

Lemma 3.2. If the regularizer $R : \mathcal{X} \rightarrow \mathbb{R}$ satisfies $|R(x) - R(x')| \leq D$ for all $x, x' \in \mathcal{X}$, then Algorithm 1 with step-size η satisfies

$$\sum_{t=1}^T g_t(x_t) - \sum_{t=1}^T g_t(x^*) \leq \sum_{t=1}^T g_t(x_t) - g_t(x_{t+1}) + \frac{D}{\eta}.$$

Lemma 3.3. If the regularizer $R : \mathcal{X} \rightarrow \mathbb{R}$ is α -strongly-convex and satisfies $|R(x) - R(x')| \leq D$ for all $x, x' \in \mathcal{X}$, then Algorithm 1 with step-size η satisfies $\|x_{t+1} - x_t\| \leq \eta \frac{L}{\alpha}$ and

$$\sum_{t=1}^T g_t(x_t) - \sum_{t=1}^T g_t(x^*) \leq \frac{D}{\eta} + \eta \frac{TL^2}{\alpha}.$$

The following lemma bounds the difference between $f_t(h_t)$ and $\mathring{f}_t(x_t)$ in terms of the p -effective memory capacity (Definition 2.3) using the Lipschitz continuity of f_t (assumption A1) and the bound on $\|x_{t+1} - x_t\|$ from Lemma 3.3.

Lemma 3.4. Consider an online convex optimization with unbounded memory problem specified by the decision space $\mathcal{X}$, history space $\mathcal{H}$, and linear operators $A : \mathcal{H} \rightarrow \mathcal{H}$ and $B : \mathcal{X} \rightarrow \mathcal{H}$. If the decisions (x_t) are generated by Algorithm 1, then

$$|f_t(h_t) - \mathring{f}_t(x_t)| \leq \eta \frac{L^2}{\alpha} H_1 \quad (5)$$

for all rounds t . When $\mathcal{H}$ is a sequence space with the $\|\cdot\|_p$ norm for $p \geq 1$, then the above bound holds with H_p instead of H_1 .

Proof. By our Lipschitz continuity assumption (assumption **A1**), we have

$$\begin{aligned} \left| f_t(h_t) - \mathring{f}_t(x_t) \right| &= \left| f_t(h_t) - f_t \left(\sum_{k=0}^{t-1} A^k B x_t \right) \right| \\ &\leq L \left\| h_t - \sum_{k=0}^{t-1} A^k B x_t \right\| \\ &\leq L \left\| \sum_{k=0}^{t-1} A^k B x_{t-k} - \sum_{k=0}^{t-1} A^k B x_t \right\| \end{aligned} \quad (6)$$

$$= L \underbrace{\left\| \sum_{k=0}^{t-1} A^k B (x_{t-k} - x_t) \right\|}_{(a)}, \quad (7)$$

where Eq. (6) follows by definition of h_t . In the general case we can bound the term (a) as

$$\begin{aligned} \left\| \sum_{k=0}^{t-1} A^k B (x_t - x_{t-k}) \right\| &\leq \sum_{k=0}^{t-1} \|A^k B\| \|x_t - x_{t-k}\| \\ &\leq \sum_{k=0}^{t-1} \|A^k B\| k \eta \frac{L}{\alpha} \\ &\leq \eta \frac{L}{\alpha} H_1, \end{aligned} \quad (8)$$

where Eq. (8) follows by Lemma 3.3. Combining this with Eq. (7) completes the proof for the general case.

When $\mathcal{H}$ is a sequence space with the $\|\cdot\|_p$ norm for $p \geq 1$, then we can bound the term (a) as

$$\begin{aligned} \left\| \sum_{k=0}^{t-1} A^k B (x_t - x_{t-k}) \right\| &= \left(\sum_{k=0}^{t-1} \|A^k (x_t - x_{t-k})\|^p \right)^{\frac{1}{p}} \\ &\leq \left(\sum_{k=0}^{t-1} \|A^k\|^p \|x_t - x_{t-k}\|^p \right)^{\frac{1}{p}} \\ &\leq \eta \frac{L}{\alpha} \left(\sum_{k=0}^{t-1} k^p \|A^k\|^p \right)^{\frac{1}{p}} \end{aligned} \quad (9)$$

$$\begin{aligned} &\leq \eta \frac{L}{\alpha} \left(\sum_{k=0}^{t-1} k^p \|A^k B\|^p \right)^{\frac{1}{p}} \\ &\leq \eta \frac{L}{\alpha} H_p, \end{aligned} \quad (10)$$

where Eq. (9) follows by Lemma 3.3 and Eq. (10) follows because $B(x) = (x, 0, 0, \dots)$. Combining this with Eq. (7) completes the proof for the case when $\mathcal{H}$ is a sequence space with the $\|\cdot\|_p$ norm for $p \geq 1$. $\blacksquare$

Proof of Theorem 3.1. As stated at the start of this section, we can write the regret as

$$\begin{aligned}
R_T(\text{FTRL}) &\leq \underbrace{\sum_{t=1}^T f_t(h_t) - \dot{f}_t(x_t)}_{(a)} + \underbrace{\sum_{t=1}^T g_t(x_t) - \min_{x \in \mathcal{X}} \sum_{t=1}^T g_t(x)}_{(b)} \\
&\leq \eta \frac{TL^2}{\alpha} H_1 + \frac{D}{\eta} + \eta \frac{TL^2}{\alpha},
\end{aligned}$$

where the last inequality follows by bounding term (a) using Lemma 3.4 and bounding term (b) using Lemma 3.3. When $\mathcal{H}$ is a sequence space with the $\|\cdot\|_p$ norm for $p \geq 1$, we can bound term (a) using Lemma 3.4 by $\eta \frac{TL^2}{\alpha} H_p$ instead. This completes the proof. $\blacksquare$

4 Applications

4.1 Special Case: OCO with Finite Memory

We start by showing how OCO with finite memory [Anava et al., 2015] is a special case of our framework.¹ Formally, let $\mathcal{X} \subseteq \mathbb{R}^d$ be closed and convex with $\|\cdot\|_{\mathcal{X}} = \|\cdot\|_2$. Let $\mathcal{H} = \mathcal{X}^m$ with $\|h\|_{\mathcal{H}} = \sqrt{\sum_{j=1}^m \|x_j\|^2}$, where $h = (x_0, \dots, x_m)$. Define the linear operators as $A : \mathcal{H} \rightarrow \mathcal{H}$ and $B : \mathcal{X} \rightarrow \mathcal{H}$ as $A(x_t, \dots, x_{t-m}) = (0, x_t, \dots, x_{t-m+1})$ and $B(x) = (x, 0, \dots, 0)$ respectively. In words, the history $h_t = (x_t, \dots, x_{t-m})$ consists of the past m decisions and gets updated as $h_{t+1} = (x_{t+1}, x_t, \dots, x_{t-m+1})$. Note that $\mathcal{H}$ is a sequence space with the $\|\cdot\|_2$ norm. Therefore, we can apply Theorem 3.1 with $p = 2$. It remains to bound H_2 .

Let $x \in \mathcal{X}$ with $\|x\|_{\mathcal{X}} = 1$ and $h \in \mathcal{H}$ with $\|h\|_{\mathcal{H}} = 1$. By definition, we have $\|Bx\|_{\mathcal{H}} = \|x\|_{\mathcal{X}} = 1$, for $k > m$ we have $\|A^k h\| = \|0\| = 0$, and for $k \leq m$ we have $\|A^k h\| \leq \|h\| = 1$. Therefore, $H_2 \leq \sqrt{\sum_{k=1}^m k^2} \leq O(m^{3/2})$. Plugging this into Theorem 3.1, we obtain a $O(m^{3/4}\sqrt{T})$ regret bound that matches existing results [Anava et al., 2015].

4.2 OCO with Infinite Memory

We now show how our framework can model OCO with *infinite* memory problems that are not modelled by existing works. Formally, let $\mathcal{X} \subseteq \mathbb{R}^d$ be closed and convex with $\|\cdot\|_{\mathcal{X}} = \|\cdot\|_2$. Let $\mathcal{H} \subseteq \mathcal{X}^{\mathbb{N}}$ consist of $h = (y_0, y_1, \dots)$ satisfying $\sum_{j=0}^{\infty} \|y_j\|^2 < \infty$ with $\|h\|_{\mathcal{H}} = \sqrt{\sum_{j=0}^{\infty} \|y_j\|^2}$. Let $\rho \in (0, 1)$. Define the linear operators $A : \mathcal{H} \rightarrow \mathcal{H}$ and $B : \mathcal{X} \rightarrow \mathcal{H}$ as $A((y_0, y_1, y_2, \dots)) = (0, \rho y_0, \rho y_1, \dots)$ and $B(x) = (x, 0, 0, \dots)$ respectively. Note that $\mathcal{H}$ is a sequence space with the $\|\cdot\|_2$ norm. Therefore, we can apply Theorem 3.1 with $p = 2$. It remains to bound H_2 .

¹The only difference is that the summation in our definition of regret starts at $t = 1$ whereas it starts at $t = m$ in the OCO with finite memory framework. However, this difference only adds $O(m)$ to the regret and m is a constant in the OCO with finite memory framework, so we can ignore it.

Let $x \in \mathcal{X}$ with $\|x\|_{\mathcal{X}} = 1$ and $h = (y_0, y_1, \dots) \in \mathcal{H}$ with $\|h\|_{\mathcal{H}} = 1$. By definition, we have that $\|Bx\|_{\mathcal{H}} = \|x\|_{\mathcal{X}} = 1$, $A^k h = (0, \dots, 0, \rho^k y_0, \rho^k y_1, \dots)$ and $\|A^k h\| = \rho^k$. Therefore,

$$H_2 \leq \sqrt{\sum_{k=1}^{\infty} k^2 \rho^{2k}} \leq O\left(\frac{\rho^4}{(1 - \rho^2)^{\frac{3}{2}}}\right).$$

A useful extension of the above is when elements of $\mathcal{X}$ are $d \times d$ matrices and the linear operator A is defined as $A((y_0, y_1, \dots)) = (0, Zy_0, Zy_1, \dots)$, where $Z \in \mathbb{R}^{d \times d}$ satisfies $\|Z\|_2 < 1$. An even more general extension of the above is the crux of our simplified regret bounds for online linear control (Section 4.4).

4.3 Online Performative Prediction

4.3.1 Background

In many applications of machine learning the algorithm's decisions influence the data distribution, for example, online labor markets [Anagnostopoulos et al., 2018, Horton, 2010], predictive policing [Lum and Isaac, 2016], on-street parking Dowling et al. [2020], Pierce and Shoup [2018], and vehicle sharing markets [Banerjee et al., 2015] to name a few. Motivated by such applications, several works have studied the problem of performative prediction, which models the data distribution as a function of the decision-maker's decision [Perdomo et al., 2020, Mendler-Dünner et al., 2020, Miller et al., 2021, Brown et al., 2022, Ray et al., 2022]. These works view the problem as a stochastic optimization problem; we refer the reader to these citations for more details. As a natural extension to existing works, we introduce an online learning variant of performative prediction with geometric decay [Ray et al., 2022] that differs from the original formulation in a few key ways.

Let $\mathcal{X} \subseteq \mathbb{R}^d$ denote the decision set with $\|x\|_2 \leq R_{\mathcal{X}}$. Let p_0 denote the initial data distribution over the instance space $\mathcal{Z}$. In each round $t \in [T]$, the learner chooses a decision $x_t \in \mathcal{X}$ and an oblivious adversary chooses a loss function $l_t : \mathcal{X} \times \mathcal{Z} \rightarrow [0, 1]$, and then the learner suffers the loss $L_t(x_t) = \mathbb{E}_{z \sim p_t} [l_t(x_t, z)]$, where $p_t = p_t(x_1, \dots, x_t)$ is the data distribution in round t . We also use the shorthand $p_t(x) = p_t(x, \dots, x)$.

The goal in our online learning setting is to minimize the difference between the algorithm's total loss and the total loss of the best fixed decision,

$$\sum_{t=1}^T \mathbb{E}_{z \sim p_t} [l_t(x_t, z)] - \min_{x \in \mathcal{X}} \sum_{t=1}^T \mathbb{E}_{z \sim p_t(x)} [l_t(x, z)],$$

as a natural generalization of performative optimality [Perdomo et al., 2020] for our online learning formulation.

We make the following assumptions about the distributions and loss functions. The data distribution satisfies $p_1 = p_0$, and for all $t \geq 1$, $p_{t+1} = \rho p_t + (1 - \rho) \mathcal{D}(x_t)$, where $\rho \in (0, 1)$ and $\mathcal{D}(x_t)$ is a distribution over $\mathcal{Z}$ that depends on the decision x_t . The distributions $\mathcal{D}(x)$ are assumed to belong to the following location-scale family: $z \sim \mathcal{D}(x)$ iff $z \sim \xi + Ax$, where $A \in \mathbb{R}^{d \times d}$ satisfies $\|A\|_2 < \infty$ and ξ is a random variable with mean μ and covariance Σ . The loss functions l_t are assumed to be convex, differentiable, and $\tilde{L}$ -Lipschitz continuous.

Our problem formulation differs from existing work in the following ways. First, we adopt an online learning perspective, whereas Ray et al. [2022] adopt a stochastic optimization perspective. Therefore, we assume that the loss functions l_t are adversarially chosen, whereas Ray et al. [2022] assume $l_t = l$ are fixed. Second, our gradient oracle (assumption A4) assumes that the dynamics ($\mathcal{D}$ and ρ) are *known*, whereas Ray et al. [2022] estimate the gradient using samples from this distribution. We believe that an appropriate extension of our framework that can deal with unknown linear operators A and B can be applied to this more difficult setting, and we leave this as future work.

Before formulating this problem in our OCO with unbounded memory framework, we state the definition of 1-Wasserstein distance that we use in our regret analysis. Informally, the 1-Wasserstein distance is a measure of the distance between two probability measures.

Definition 4.1 (1-Wasserstein Distance). Let $(\mathcal{Z}, d)$ be a metric space. Let $\mathbb{P}(\mathcal{Z})$ denote the set of Radon probability measures ν on $\mathcal{Z}$ with finite first moment. That is, there exists $z' \in \mathcal{Z}$ such that $\mathbb{E}_{z \sim \nu}[d(z, z')] < \infty$. The 1-Wasserstein distance between two probability measures $\nu, \nu' \in \mathbb{P}(\mathcal{Z})$ is defined as

$$W_1(\nu, \nu') = \sup\{\mathbb{E}_{z \sim \nu}[f(z)] - \mathbb{E}_{z \sim \nu'}[f(z)]\},$$

where the supremum is taken over all 1-Lipschitz continuous functions $f : \mathcal{Z} \rightarrow \mathbb{R}$.

4.3.2 Formulation as OCO with Unbounded Memory

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be closed and convex with $\|\cdot\|_{\mathcal{X}} = \|\cdot\|_2$. Let $\mathcal{H} \subseteq \mathcal{X}^{\mathbb{N}}$ consist of sequences $h = (x_0, \rho x_1, \rho^2 x_2, \dots)$, where $\rho \in (0, 1)$ and $x_k \in \mathcal{X}$ satisfy $\sum_{k=0}^{\infty} \|x_k\|^2 < \infty$, with $\|h\|_{\mathcal{H}} = \sqrt{\sum_{k=0}^{\infty} \rho^{2k} \|x_k\|^2}$.

Define the linear operators $A : \mathcal{H} \rightarrow \mathcal{H}$ and $B : \mathcal{X} \rightarrow \mathcal{H}$ as $A((y_0, y_1, y_2, \dots)) = (0, \rho y_0, \rho y_1, \dots)$ and $B(x) = (x, 0, 0, \dots)$ respectively. Note that $\mathcal{H}$ is a sequence space with the $\|\cdot\|_2$ norm.

By definition, given a sequence of decisions $(x_1, \dots, x_t)$, the history at the end of round t is given by $h_t = (x_t, \rho x_{t-1}, \rho^2 x_{t-2}, \dots, \rho^t x_1, 0, \dots)$. Define the functions $f_t : \mathcal{H} \rightarrow [0, 1]$ by $f_t(h_t) = \mathbb{E}_{z \sim p_t(x_1, \dots, x_t)}[l_t(x_t, z)]$, where $p_t(x_1, \dots, x_t)$ is the distribution that satisfies $z \sim p_t$ iff

$$z \sim \sum_{k=1}^{t-1} (1 - \rho) \rho^{k-1} (\xi + A x_{t-k}) + \rho^t p_0. \quad (11)$$

Note that the distribution p_t is a function of the history h_t . The above follows from the recursive definition of p_t and parametric assumption about $\mathcal{D}(x)$.

With the above formulation and definition of f_t , the original goal of minimizing the difference between the algorithm's total loss and the total loss of the best fixed decision is equivalent to minimizing the regret,

$$\sum_{t=1}^T f_t(h_t) - \min_{x \in \mathcal{X}} \sum_{t=1}^T \mathring{f}_t(x).$$

4.3.3 Regret Analysis

In order to apply our regret bounds, we must bound H_2 , the Lipschitz continuity constant (assumption **A1**), and the norm of the gradients (assumption **A2**).

Lemma 4.1. *We have $H_2 \leq O\left(\frac{\rho^4}{(1-\rho^2)^{\frac{3}{2}}}\right) < \infty$.*

Proof. Note that $\|B\| = 1$ and $\|A^k\| \leq \rho^k$. Therefore,

$$H_2 = \sqrt{\sum_{k=0}^{\infty} k^2 \|A^k B\|^2} \leq \sqrt{\sum_{k=0}^{\infty} k^2 \rho^{2k}} \leq O\left(\frac{\rho^4}{(1-\rho^2)^{\frac{3}{2}}}\right). \quad \blacksquare$$

Lemma 4.2. *We have the following bound on the Lipschitz continuity constant: $L \leq \tilde{L}^{\frac{1-\rho}{\rho}} \|A\|_2$.*

Proof. Let $(x_0, \dots, x_t)$ and $(\tilde{x}_1, \dots, \tilde{x}_t)$ be two sequences of decisions, where $x_k, \tilde{x}_k \in \mathcal{X}$. Let h_t and $\tilde{h}_t$ be the corresponding histories, and p_t and $\tilde{p}_t$ be the corresponding distributions at the end of round t . We have

$$\begin{aligned} & \left| f_t(h_t) - f_t(\tilde{h}_t) \right| \\ &= \left| \mathbb{E}_{z \sim p_t} [l_t(x_t, z)] - \mathbb{E}_{z \sim \tilde{p}_t} [l_t(\tilde{x}_t, z)] \right| \\ &= \left| \mathbb{E}_{z \sim p_t} [l_t(x_t, z)] - \mathbb{E}_{z \sim p_t} [l_t(\tilde{x}_t, z)] \right. \\ & \quad \left. + \mathbb{E}_{z \sim p_t} [l_t(\tilde{x}_t, z)] - \mathbb{E}_{z \sim \tilde{p}_t} [l_t(\tilde{x}_t, z)] \right| \\ &\leq \tilde{L} \|x_t - \tilde{x}_t\|_2 + \tilde{L} W_1(p_t, \tilde{p}_t), \end{aligned}$$

where the last inequality follows from the assumptions about the functions l_t and the definition of the Wasserstein distance W_1 . By definition of p_t Eq. (11), we have that

$$\begin{aligned} W_1(p_t, \tilde{p}_t) &\leq \sum_{k=1}^{t-1} \frac{1-\rho}{\rho} \rho^k \|A\|_2 \|x_{t-k} - \tilde{x}_{t-k}\|_2 \\ &\leq \frac{1-\rho}{\rho} \|A\|_2 \|h_t - \tilde{h}_t\|_2, \end{aligned}$$

where the last inequality follows from the definition of $\|\cdot\|_{\mathcal{H}}$. Therefore, $L \leq \tilde{L}^{\frac{1-\rho}{\rho}} \|A\|_2$. $\blacksquare$

Lemma 4.3. *We have the following bound on the norm of the gradient: $\|\nabla f_t^\circ(x)\| \leq \tilde{L}(1 + \|A\|_2)$.*

Proof. We have [Ray et al., 2022]

$$\nabla f_t^\circ(x) = \mathbb{E}_{z \sim p_t} [\nabla_x l_t(x, z) + (1 - \rho^t) A^\top \nabla_z l_t(x, z)].$$

The results follows because the functions l_t are assumed to be $\tilde{L}$ -Lipschitz continuous in (x, z) . $\blacksquare$

Theorem 4.1. *For the online performative prediction problem, we have*

$$R_T(\text{FTRL}) \leq O\left(R_{\mathcal{X}} \tilde{L} L' \frac{1-\rho}{\rho} \sqrt{1 + \frac{\rho^4}{(1-\rho^2)^{\frac{3}{2}}}} \sqrt{T}\right),$$

where $L' = \max\{1 + \|A\|_2, \frac{1-\rho}{\rho} \|A\|_2\}$.

Proof. Choosing the $R(\cdot) = \frac{1}{2} \|\cdot\|_{\mathcal{X}}^2$ as the 1-strongly convex regularizer, we have that for all $x, x' \in \mathcal{X}$,

$$|R(x) - R(x')| \leq R_{\mathcal{X}}^2 = D.$$

Recall the regret bound from Theorem 3.1. Using the above bound on D and using Lemmas 4.1 to 4.3 to bound the other constants, we obtain the result. $\blacksquare$

4.4 Online Linear Control

4.4.1 Background

Now we use our framework to simplify regret bound derivations for online linear control [Agarwal et al., 2019b]. Existing works model it (and its extensions, such as, Agarwal et al. [2019c], Hazan et al. [2020], Li et al. [2021] to name a few) as an OCO with finite memory problem. This necessitates a lot of truncation and error analysis that constitutes the bulk of existing proofs. On the other hand, this problem fits very naturally into our framework because it is inherently an OCO with unbounded memory problem. We refer the reader to existing literature [Agarwal et al., 2019b] for more details and background on online linear control. Here, we introduce the basic mathematical setup of the problem.

Let $\mathcal{S} \subseteq \mathbb{R}^{d_s}$ denote the state space, $\mathcal{U} \subseteq \mathbb{R}^{d_u}$ denote the control space, and s_t and u_t denote the state and control at time t with s_0 being the initial state. The system evolves according to the linear dynamics $s_{t+1} = F s_t + G u_t + w_t$, where $F \in \mathbb{R}^{d_s \times d_s}$, $G \in \mathbb{R}^{d_s \times d_u}$ are matrices satisfying $\|F\|_2, \|G\|_2 \leq \kappa$ and $w_t \in \mathbb{R}^{d_s}$ is an adversarially chosen “disturbance” with $\|w\|_{\infty} \leq W$. We assume, without loss of generality and for notational convenience, that $d_s = d_u = d$ and also define $w_{-1} = s_0$.

For $t = 0, \dots, T-1$, let $c_t : \mathcal{S} \times \mathcal{U} \rightarrow [0, 1]$ be convex loss functions chosen by an oblivious adversary. The functions c_t are assumed to satisfy the following Lipschitz condition: if $\|s\|_2, \|u\|_2 \leq D_{\mathcal{X}}$, then $\|\nabla_s c_t(s, u)\|, \|\nabla_u c_t(s, u)\| \leq \tilde{L} D_{\mathcal{X}}$. The goal in online linear control is to choose a sequence of controls u_t to minimize the following regret

$$R_T(\Pi) = \sum_{t=0}^{T-1} c_t(s_t, u_t) - \min_{\pi^* \in \Pi} \sum_{t=0}^{T-1} c_t(s_t^{\pi^*}, u_t^{\pi^*}), \quad (12)$$

where s_t evolves according to linear dynamics as stated above, Π denotes an arbitrary policy class, and $s_t^{\pi^*}, u_t^{\pi^*}$ denote the state and control at time t when the controls are chosen according to the policy π^* .

Like previous work [Agarwal et al., 2019b] we focus on the policy class of (κ, ρ) -strongly stable linear controllers, $\mathcal{K}$, where $K \in \mathcal{K}$ satisfies $F - GK = HLH^{-1}$ with $\|K\|_2, \|H\|_2, \|H^{-1}\|_2 \leq \kappa$ and $\|L\|_2 = \rho < 1$. Given such a controller, the controls are chosen as linear functions of

the current state, i.e., $u_t = -Ks_t$. The motivation for considering this policy class comes from the fact that for an infinite horizon problem with quadratic costs, the optimal controller is a fixed linear controller.

Unfortunately, parameterizing u_t directly with a linear controller as $u_t = -Ks_t$ leads to a non-convex problem because s_t is a non-linear function of K . To see this, suppose the disturbances are equal to zero for simplicity. Then, $s_t = \tilde{F}^t s_0$, where we use $\tilde{F} = (F - GK)$ throughout the paper for notational convenience. An alternative parameterization is the disturbance-action controller.

Definition 4.2. A disturbance-action controller (K, M) is specified by a *fixed* matrix $K \in \mathcal{K}$ and parameters $M = (M^{[1]}, M^{[2]}, \dots)$, where each $M^{[s]} \in \mathbb{R}^{d \times d}$ satisfies $\|M^{[i]}\|_2 \leq \kappa^4 \rho^i$. The control in round k is chosen as

$$u_k = -Ks_k + \sum_{s=1}^{k+1} M^{[s]} w_{k-s}.$$

The class of such disturbance-action controllers has two important properties. First, given any linear controllers $K, K^* \in \mathcal{K}$, K^* can be expressed as a disturbance-action controller (K, M) , where $M = (M^{[1]}, \dots, M^{[T+1]}, 0, \dots)$ [Agarwal et al., 2019a, Section 16.5]. In other words, the class of disturbance-action controllers is a superset of the class of (κ, ρ) -strongly-stable linear controllers. An important consequence is that it suffices to compute regret against the class of disturbance-action controllers instead of $\mathcal{K}$. Second, let $M_0, \dots, M_t$ denote the parameters in round t , where each $M_k = (M_k^{[1]}, M_k^{[2]}, \dots)$ as defined above. A simple inductive argument shows that the state and control at time t are *linear* functions of the parameters $(M_0, \dots, M_t)$,

$$s_t = \tilde{F}^t s_0 + \sum_{k=0}^{t-1} \sum_{s=1}^{k+1} \tilde{F}^{t-k-1} G M_k^{[s]} w_{k-s} + w_{t-1}, \quad (13)$$

$$u_t = -Ks_t + \sum_{s=1}^{t+1} M_t^{[s]} w_{t-s}. \quad (14)$$

This implies that c_t is convex in the parameters.

A similar parameterization was first considered for online linear control in Agarwal et al. [2019b] and is based on similar ideas in control theory, e.g., the classic Youla parameterization [Youla et al., 1976, Kučera, 1975] and a more recent generalization called system level synthesis framework [Wang et al., 2019, Anderson et al., 2019].

For notational convenience, let $(M^{[s]})$ and (Y_k) denote the sequences $(M^{[1]}, M^{[2]}, \dots)$ and $(Y_0, Y_1, \dots)$ respectively.

Let

$$\mathcal{X} = \{M = (M^{[s]}) : M^{[s]} \in \mathbb{R}^{d \times d}, \|M^{[s]}\|_2 \leq \kappa^4 \rho^s\}$$

with $\|M\|_{\mathcal{X}} = \sqrt{\sum_{s=1}^{\infty} \rho^{-s} \|M^{[s]}\|_F^2}$. Note that the norm on $\mathcal{X}$ is induced by the weighted inner product $\langle M_1, M_2 \rangle = \sum_{s=1}^{\infty} \rho^{-s} \|M_1^{[s]}\|_F \|M_2^{[s]}\|_F$. Note that even though the weights ρ^{-s} increase exponentially, the norms of the matrices decrease exponentially as well.

Let $\mathcal{H}$ consist of sequences $h = (Y_k)$, where $Y_0 \in \mathcal{X}$ and $Y_k = \tilde{F}^{k-1} G X_k$ for $X_k \in \mathcal{X}, k \geq 1$, with $\|h\|_{\mathcal{H}} = \|Y_0\| + \sum_{k=1}^{\infty} \|\tilde{F}^{k-1} G\|_2 \|X_k\|_{\mathcal{X}}$.

Define the linear operators $A : \mathcal{H} \rightarrow \mathcal{H}$ and $B : \mathcal{X} \rightarrow \mathcal{H}$ as $A((Y_0, Y_1, \dots)) = (0, GY_0, \tilde{F}Y_1, \tilde{F}Y_2, \dots)$ and $B(M) = (M, 0, 0, \dots)$ respectively.

By definition, given a sequence of decisions $(M_0, \dots, M_t)$, the history at the end of round t is given by

$$h_t = (M_t, GM_{t-1}, \tilde{F}GM_{t-2}, \dots, \tilde{F}^{t-1}GM_0, 0, \dots).$$

Define the functions $f_t : \mathcal{H} \rightarrow [0, 1]$ by $f_t(h) = c_t(s, u)$, where s and u are the state and control determined by the history as in Eqs. (13) and (14). Since the state and control are linear functions of the history and c_t is convex, this implies that f_t is convex.

With the above formulation and the fact that the class of disturbance-action controllers is a superset of the class of (κ, ρ) -strongly-stable controllers, we have that $R_T(\mathcal{K})$ (Eq. (12)) is at most

$$\sum_{t=0}^{T-1} f_t(h_t) - \min_{M \in \mathcal{X}} \sum_{t=0}^{T-1} \hat{f}_t(M).$$

4.4.2 Regret Analysis

Even though online linear control is inherently an OCO with unbounded memory problem, existing works [Agarwal et al., 2019b] reduce it to OCO with *finite* memory and must analyze the error between the finite memory reduction and the inherent unbounded memory nature of the problem. Furthermore, they consider a restricted class of disturbance-action controllers, where each parameter is truncated to be a sequence of fixed length. Consequently, this class is no longer a superset of the class of (κ, ρ) -strongly-stable linear controllers, and introduces an error term that must be analyzed. In contrast, our regret analysis is greatly simplified because we reduce directly to our OCO with *unbounded* memory framework, which allows us to skip these error analysis steps entirely.

In order to apply our regret bounds, we must bound H_1 , the Lipschitz continuity constant (assumption A1), and the norm of the gradients (assumption A2). We note that existing works [Agarwal et al., 2019b] also bound the norm of the gradients. However, they only bound the *coordinate-wise Lipschitz* continuity constant, which is weaker than the *Lipschitz* continuity constant that we bound.

Lemma 4.4. *We have $H_1 \leq 2\frac{\kappa^4\rho}{1-\rho} < \infty$.*

Proof. Note that $\|B\| = 1$ and $\|A^k\| \leq \kappa^4\rho^{k-1}$. Therefore,

$$H_1 = \sum_{k=0}^{\infty} k\|A^k B\| \leq \kappa^4 \sum_{k=0}^{\infty} k\rho^{k-1} \leq 2\frac{\kappa^4\rho}{1-\rho}. \quad \blacksquare$$

Lemma 4.5. *We have $\|s_t\|, \|u_t\| \leq D_{\mathcal{X}} = O\left(\sqrt{d}W\kappa^8\frac{1}{(1-\rho)^2}\right)$.*

Proof. We can bound the norms of the states and controls as

$$\begin{aligned}
\|s_t\|_2 &\leq \left\| \tilde{F}^t s_0 + \sum_{k=0}^{t-1} \sum_{s=1}^{k+1} \tilde{F}^{t-k-1} G M_k^{[s]} w_{k-s} + w_{t-1} \right\|_2 \\
&\leq \kappa^2 \rho^t + \sum_{k=0}^{t-1} \sum_{s=1}^{k+1} \kappa^2 \rho^{t-k-1} \kappa \kappa^4 \rho^s \sqrt{d} W + \sqrt{d} W \\
&\leq \kappa^2 + \sqrt{d} W + \sqrt{d} W \kappa^7 \frac{1}{(1-\rho)^2} \\
&\leq O\left(\sqrt{d} W \kappa^7 \frac{1}{(1-\rho)^2}\right). \\
\|u_t\|_2 &\leq \left\| K s_t + \sum_{s=1}^{t+1} M_t^{[s]} w_{t-s} \right\|_2 \\
&\leq O\left(\sqrt{d} W \kappa^8 \frac{1}{(1-\rho)^2}\right) + \sum_{s=1}^{t+1} \kappa^4 \rho^s \sqrt{d} W \\
&\leq O\left(\sqrt{d} W \kappa^8 \frac{1}{(1-\rho)^2}\right).
\end{aligned}$$

■

Lemma 4.6. *We have the following bound on the Lipschitz continuity constant: $L \leq \tilde{L} D_{\mathcal{X}} \frac{\sqrt{d} W \kappa}{\sqrt{1-\rho}}$.*

Proof. Let $(M_0, \dots, M_t)$ and $(\widetilde{M}_0, \dots, \widetilde{M}_t)$ be two sequences of decisions, where M_k and $\widetilde{M}_k \in \mathcal{X}$. Let h_t and $\tilde{h}_t$ be the corresponding histories, and (s_t, u_t) and $(\tilde{s}_t, \tilde{u}_t)$ be the corresponding state-control pairs at the end of round t . We have

$$\begin{aligned}
\left| f_t(h_t) - f_t(\tilde{h}_t) \right| &= |c_t(s_t, u_t) - c_t(\tilde{s}_t, \tilde{u}_t)| \\
&\leq \tilde{L} D_{\mathcal{X}} \max \{ \|s_t - \tilde{s}_t\|_2, \|u_t - \tilde{u}_t\|_2 \},
\end{aligned}$$

where the last inequality follows from the assumptions about the functions c_t and the bound $\|s_t\|_2, \|u_t\|_2 \leq D_{\mathcal{X}}$. For $k = 0, \dots, t-1$, define $Z_k^{[s]} = \tilde{F}^{t-k-1} G(M_k^{[s]} - \widetilde{M}_k^{[s]})$. Using Eq. (13),

observe that

$$\begin{aligned}
& \|s_t - \tilde{s}_t\|_2 \\
&= \left\| \sum_{k=0}^{t-1} \sum_{s=1}^{k+1} Z_k^{[s]} w_{k-s} \right\|_2 \\
&\leq \sum_{k=0}^{t-1} \sum_{s=1}^{k+1} \left\| Z_k^{[s]} w_{k-s} \right\|_2 \\
&= \sum_{k=0}^{t-1} \sum_{s=1}^{k+1} \left\| \rho^{-\frac{s}{2}} Z_k^{[s]} \rho^{\frac{s}{2}} w_{k-s} \right\|_2 \\
&\leq \sum_{k=0}^{t-1} \sum_{s=1}^{k+1} \left\| \rho^{-\frac{s}{2}} Z_k^{[s]} \right\|_2 \left\| \rho^{\frac{s}{2}} w_{k-s} \right\|_2 \\
&\leq \sum_{k=0}^{t-1} \sqrt{\sum_{s=1}^{k+1} \left\| \rho^{-\frac{s}{2}} Z_k^{[s]} \right\|_2^2} \sqrt{\sum_{s=1}^{k+1} \left\| \rho^{\frac{s}{2}} w_{k-s} \right\|_2^2} \tag{15}
\end{aligned}$$

$$\begin{aligned}
&\leq \frac{\sqrt{d}W}{\sqrt{1-\rho}} \sum_{k=0}^{t-1} \sqrt{\sum_{s=1}^{k+1} \rho^{-s} \left\| Z_k^{[s]} \right\|_2^2} \\
&= \frac{\sqrt{d}W}{\sqrt{1-\rho}} \sum_{k=0}^{t-1} \sqrt{\sum_{s=1}^{k+1} \rho^{-s} \left\| \tilde{F}^{t-k-1} G(M_k^{[s]} - \tilde{M}_k^{[s]}) \right\|_2^2} \\
&\leq \frac{\sqrt{d}W}{\sqrt{1-\rho}} \sum_{k=0}^{t-1} \|\tilde{F}^{t-k-1} G\|_2 \sqrt{\sum_{s=1}^{k+1} \rho^{-s} \left\| (M_k^{[s]} - \tilde{M}_k^{[s]}) \right\|_2^2} \\
&\leq \frac{\sqrt{d}W}{\sqrt{1-\rho}} \sum_{k=0}^{t-1} \|\tilde{F}^{t-k-1} G\|_2 \sqrt{\sum_{s=1}^{k+1} \rho^{-s} \left\| (M_k^{[s]} - \tilde{M}_k^{[s]}) \right\|_F^2} \tag{16} \\
&= \frac{\sqrt{d}W}{\sqrt{1-\rho}} \|h_t - \tilde{h}_t\|_{\mathcal{H}},
\end{aligned}$$

where Eq. (15) follows from the Cauchy-Schwarz inequality, Eq. (16) follows because $\|\cdot\|_2 \leq \|\cdot\|_F$, and the last equality follows by definition of $\|\cdot\|_{\mathcal{X}}$ and $\|\cdot\|_{\mathcal{H}}$. Similarly,

$$\begin{aligned}
& \|u_t - \tilde{u}_t\|_2 \\
&= \left\| K(s_t - \tilde{s}_t) + \sum_{s=1}^{t+1} (M_t^{[s]} - \tilde{M}_t^{[s]}) w_{t-s} \right\|_2 \\
&\leq \frac{\sqrt{d}W\kappa}{\sqrt{1-\rho}} \|h_t - \tilde{h}_t\|_{\mathcal{H}}.
\end{aligned}$$

This completes the proof. ■

Lemma 4.7. *We have the following bound on the norm of the gradient: $\left\| \nabla \mathring{f}_t(M) \right\| \leq \tilde{L} D_{\mathcal{X}} \frac{d^{\frac{3}{2}} W \kappa^4}{(1-\rho)^2}$.*

Proof. Note that $M = (M^{[s]}) \in \mathcal{X}$. We have

$$\left\| \nabla \mathring{f}_t(M) \right\| \leq \tilde{L} D_{\mathcal{X}} \{ \|\nabla_M s_t\|, \|\nabla_M u_t\| \},$$

where the inequality follows from the assumptions about the functions c_t and the bound $\|s_t\|_2, \|u_t\|_2 \leq D_{\mathcal{X}}$. Here, s_t and u_t are functions of M because they are functions of the history h_t , which is history generated by playing $M \in \mathcal{X}$ in rounds $1, \dots, t$.

Note that the inner product on $\mathcal{X}$ is a non-standard inner product. Therefore, the gradient is *not* simply the “vector” (more precisely, in this case, a 3-dimensional tensor) of partial derivatives. In order to simplify the gradient computation, we first perform a reparameterization. Denote the reparameterization as $\widetilde{M} = (\widetilde{M}^{[s]})$, where $\widetilde{M}^{[s]} = \rho^{-\frac{s}{2}} M^{[s]}$. Note that the standard inner product with respect to $\widetilde{M}$ is equal to the non-standard inner product with respect to M . Therefore, in order to compute the gradient, it suffices to compute the partial derivatives with respect to $\widetilde{M}$.

Consider the partial derivative of s_t and u_t with respect to $M_{p,q}^{[s]}$, where $M_{p,q}^{[s]}$ denotes the (p, q) -th entry of $M^{[s]}$. From the definition of s_t in terms of the history (Eq. (13)), we have

$$\begin{aligned} & \left\| \frac{\partial}{\partial \widetilde{M}_{p,q}^{[s]}} s_t \right\| \\ &= \left\| \frac{\partial}{\partial \widetilde{M}_{p,q}^{[s]}} \sum_{k=0}^{t-1} \widetilde{F}^{t-k-1} G M^{[s]} w_{k-s} \right\| \\ &= \left\| \frac{\partial}{\partial \widetilde{M}_{p,q}^{[s]}} \sum_{k=0}^{t-1} \widetilde{F}^{t-k-1} G \rho^{\frac{s}{2}} \widetilde{M}^{[s]} w_{k-s} \right\| \\ &\leq \sqrt{d} W \sum_{k=0}^{t-1} \kappa^2 \rho^{t-k-1} \kappa \rho^{\frac{s}{2}} \\ &\leq \frac{\sqrt{d} W \kappa^3}{1-\rho} \rho^{\frac{s}{2}}. \end{aligned}$$

The first equality follows because the the partial derivative of $\widetilde{M}_{p,q}^{[s]}$ with respect to $M^{[s']}$ for $s \neq s'$ is zero. Summing over all s, p, q , we get that

$$\begin{aligned} \|\nabla s_t\| &\leq \sqrt{\sum_{s,p,q} \left(\frac{\sqrt{d} W \kappa^3}{1-\rho} \rho^{\frac{s}{2}} \right)^2} \\ &\leq \frac{d^{\frac{3}{2}} W \kappa^3}{(1-\rho)^2}. \end{aligned}$$

Similarly,

$$\|\nabla u_t\| \leq \frac{d^{\frac{3}{2}} W \kappa^4}{(1-\rho)^2}.$$

This completes the proof. ■

Theorem 4.2. *For the online linear control problem, we have*

$$R_T(\text{FTRL}) \leq O\left(\tilde{L}d^2W^2\kappa^{18}\frac{\rho}{(1-\rho)^5}\sqrt{T}\right).$$

Proof. Choosing the $R(\cdot) = \frac{1}{2}\|\cdot\|_{\mathcal{X}}^2$ as the 1-strongly convex regularizer, we have that for all $M, M' \in \mathcal{X}$,

$$|R(M) - R(M')| \leq \sum_{s=1}^{\infty} \rho^{-s} \kappa^8 \rho^{2s} \leq \frac{\kappa^8}{1-\rho} = D.$$

Recall the regret bound from Theorem 3.1. Using the above bound on D and using Lemmas 4.4 to 4.7 to bound the other constants, we obtain the result. ■

The polynomial dependence of the above bound on the problem parameters is comparable to existing work. Our regret bound for the online linear control problem (Theorem 4.2 is

$$O\left(\tilde{L}d^2W^2\kappa^{18}\frac{\rho}{(1-\rho)^5}\sqrt{T}\right). \quad (17)$$

Our regret bound matches that of Agarwal et al. [2019b] in terms of dependence on $\tilde{L}, W$, and T . However, there are three differences.

1. Our regret bound depends on d as d^2 , whereas the regret bound of Agarwal et al. [2019b] depends on d as $d^{\frac{3}{2}}$. This is because we assume $\|w\|_{\infty} \leq W$, instead of $\|w\|_2 \leq W$ (as they do).
2. Our regret bound depends on κ as κ^{18} , whereas the regret bound of Agarwal et al. [2019b] depends on κ as κ^{20} . (They use different notation, but the κ_B in their work is equal to κ_G in our work, which we assume equal to κ without loss of generality and for notational convenience.)
3. Our regret bound depends on $\rho = 1 - \gamma$ as $\gamma^{-5}(1 - \gamma)$, whereas the regret bound of Agarwal et al. [2019b] depends on $\rho = 1 - \gamma$ as $\gamma^{-4.5}(1 - \kappa^2(1 - \gamma)^{\frac{\kappa^2}{\gamma} \log T + 1})^{-1}$.

5 Conclusion

In this paper we introduced a generalization of the OCO framework, “Online Convex Optimization with Unbounded Memory”, that captures long-term dependence of the loss in the current round on past decisions. We introduced the notion of p -effective memory capacity, H_p , that quantifies the maximum influence of past decisions on present losses. We proved a $O(\sqrt{H_1 T})$ policy regret bound that is optimal in the time horizon T and a stronger $O(\sqrt{H_p T})$ policy regret bound when the Banach space $\mathcal{H}$ is a sequence space with the $\|\cdot\|_p$ norm, and showed that our results match existing ones for the OCO with finite memory framework. Finally, we showed the broad applicability of our framework to derive regret

bounds, and to simplify existing regret bound derivations, for a variety of online learning problems including an online variant of performative prediction online linear control.

There are a number of directions for future research. A natural follow-up to this work would be consider the feedback model where the learner has access to only a value oracle and not a gradient oracle, i.e., *bandit* convex optimization with unbounded memory. Similar extensions have been considered in the OCO and OCO with finite memory literature (e.g., Hazan and Li [2016], Bubeck et al. [2021], Zhao et al. [2021], Gradu et al. [2020], Cassel and Koren [2020]). In a similar vein, one can consider other extensions as in the OCO and OCO with finite memory literature, such as, incorporating switching costs [Shi et al., 2020], non-stationarity [Zhao et al., 2022], etc. Another follow-up would be to consider the case when the linear operators A and B are unknown, and the learner must estimate them based on past decisions and feedback. An interesting, but challenging, direction is to investigate the optimality of the regret bounds on the p -effective memory capacity H_p . Lower bounds are unknown even in the much simpler case of OCO with finite memory. Finally, it is tempting to think about a version where the history is a *nonlinear*, but decaying, function of the past decisions. The obvious challenge is that the nonlinearity would lead to non-convex losses. It is unclear how to deal with such issues, e.g., considering restricted classes of nonlinearities for which the OCO with unbounded memory perspective is still relevant [Zhang et al., 2015], considering different problem formulations such as online non-convex learning [Gao et al., 2018, Suggala and Netrapalli, 2020], etc.

Optimization and machine learning are at the core of many automated decision-making systems. In a variety of domains—such as robotics, personalized search, and social media—these decisions impact the future, and thus the perspective of online learning and optimization with memory is relevant. While the use of such automated technologies can lead to negative social impact, we do not anticipate that such impacts will arise directly from our theoretical work.

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References

- Jacob D. Abernethy, Elad Hazan, and Alexander Rakhlin. Competing in the dark: An efficient algorithm for bandit linear optimization. In Rocco A. Servedio and Tong Zhang, editors, *21st Annual Conference on Learning Theory - COLT 2008, Helsinki, Finland, July 9-12, 2008*, pages 263–274. Omnipress, 2008. URL <http://colt2008.cs.helsinki.fi/papers/127-Abernethy.pdf>.
- Alekh Agarwal, Nan Jiang, Sham M Kakade, and Wen Sun. Reinforcement learning: Theory and algorithms. 2019a.
- Naman Agarwal, Brian Bullins, Elad Hazan, Sham M. Kakade, and Karan Singh. Online

- control with adversarial disturbances. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, volume 97 of *Proceedings of Machine Learning Research*, pages 111–119. PMLR, 2019b.
- Naman Agarwal, Elad Hazan, and Karan Singh. Logarithmic regret for online control. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett, editors, *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pages 10175–10184, 2019c. URL <https://proceedings.neurips.cc/paper/2019/hash/78719f11fa2df9917de3110133506521-Abstract.html>.
- Aris Anagnostopoulos, Carlos Castillo, Adriano Fazzzone, Stefano Leonardi, and Evimaria Terzi. Algorithms for hiring and outsourcing in the online labor market. In Yike Guo and Faisal Farooq, editors, *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, KDD 2018, London, UK, August 19-23, 2018*, pages 1109–1118. ACM, 2018. doi: 10.1145/3219819.3220056. URL <https://doi.org/10.1145/3219819.3220056>.
- Oren Anava, Elad Hazan, and Shie Mannor. Online learning for adversaries with memory: Price of past mistakes. In Corinna Cortes, Neil D. Lawrence, Daniel D. Lee, Masashi Sugiyama, and Roman Garnett, editors, *Advances in Neural Information Processing Systems 28: Annual Conference on Neural Information Processing Systems 2015, December 7-12, 2015, Montreal, Quebec, Canada*, pages 784–792, 2015.
- James Anderson, John C Doyle, Steven H Low, and Nikolai Matni. System level synthesis. *Annual Reviews in Control*, 47:364–393, 2019.
- Baruch Awerbuch and Robert Kleinberg. Online linear optimization and adaptive routing. *J. Comput. Syst. Sci.*, 74(1):97–114, 2008. doi: 10.1016/j.jcss.2007.04.016. URL <https://doi.org/10.1016/j.jcss.2007.04.016>.
- Siddhartha Banerjee, Carlos Riquelme, and Ramesh Johari. Pricing in ride-share platforms: A queueing-theoretic approach. *Available at SSRN 2568258*, 2015.
- Kush Bhatia and Karthik Sridharan. Online learning with dynamics: A minimax perspective. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/abb451a12cf1a9d93292e81f0d4fdd7a-Abstract.html>.
- Gavin Brown, Shlomi Hod, and Iden Kalemaj. Performative prediction in a stateful world. In Gustau Camps-Valls, Francisco J. R. Ruiz, and Isabel Valera, editors, *International Conference on Artificial Intelligence and Statistics, AISTATS 2022, 28-30 March 2022, Virtual Event*, volume 151 of *Proceedings of Machine Learning Research*, pages 6045–6061. PMLR, 2022. URL <https://proceedings.mlr.press/v151/brown22a.html>.

- Sébastien Bubeck, Ronen Eldan, and Yin Tat Lee. Kernel-based methods for bandit convex optimization. *J. ACM*, 68(4):25:1–25:35, 2021. doi: 10.1145/3453721. URL <https://doi.org/10.1145/3453721>.
- Asaf B. Cassel and Tomer Koren. Bandit linear control. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/64a08e5f1e6c39faeb90108c430eb120-Abstract.html>.
- Nicolò Cesa-Bianchi and Gábor Lugosi. *Prediction, learning, and games*. Cambridge University Press, 2006. ISBN 978-0-521-84108-5. doi: 10.1017/CBO9780511546921. URL <https://doi.org/10.1017/CBO9780511546921>.
- Thomas M Cover. Universal portfolios. *Mathematical finance*, 1(1):1–29, 1991.
- Ofer Dekel, Ambuj Tewari, and Raman Arora. Online bandit learning against an adaptive adversary: from regret to policy regret. In *Proceedings of the 29th International Conference on Machine Learning, ICML 2012, Edinburgh, Scotland, UK, June 26 - July 1, 2012*. icml.cc / Omnipress, 2012. URL <http://icml.cc/2012/papers/749.pdf>.
- Chase P. Dowling, Lillian J. Ratliff, and Baosen Zhang. Modeling curbside parking as a network of finite capacity queues. *IEEE Trans. Intell. Transp. Syst.*, 21(3):1011–1022, 2020. doi: 10.1109/TITS.2019.2900642. URL <https://doi.org/10.1109/TITS.2019.2900642>.
- Xiand Gao, Xiaobo Li, and Shuzhong Zhang. Online learning with non-convex losses and non-stationary regret. In Amos J. Storkey and Fernando Pérez-Cruz, editors, *International Conference on Artificial Intelligence and Statistics, AISTATS 2018, 9-11 April 2018, Playa Blanca, Lanzarote, Canary Islands, Spain*, volume 84 of *Proceedings of Machine Learning Research*, pages 235–243. PMLR, 2018. URL <http://proceedings.mlr.press/v84/gao18a.html>.
- Paula Gradu, John Hallman, and Elad Hazan. Non-stochastic control with bandit feedback. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/7a1d9028a78f418cb8f01909a348d9b2-Abstract.html>.
- Elad Hazan. Introduction to online convex optimization. *CoRR*, abs/1909.05207, 2019.
- Elad Hazan and Yuanzhi Li. An optimal algorithm for bandit convex optimization. *CoRR*, abs/1603.04350, 2016. URL <http://arxiv.org/abs/1603.04350>.

- Elad Hazan, Sham M. Kakade, and Karan Singh. The nonstochastic control problem. In Aryeh Kontorovich and Gergely Neu, editors, *Algorithmic Learning Theory, ALT 2020, 8-11 February 2020, San Diego, CA, USA*, volume 117 of *Proceedings of Machine Learning Research*, pages 408–421. PMLR, 2020. URL <http://proceedings.mlr.press/v117/hazan20a.html>.
- John J Horton. Online labor markets. In *International workshop on internet and network economics*, pages 515–522. Springer, 2010.
- Vladimír Kučera. Stability of discrete linear feedback systems. *IFAC Proceedings Volumes*, 8(1):573–578, 1975.
- Yingying Li, Subhro Das, and Na Li. Online optimal control with affine constraints. In *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021, Thirty-Third Conference on Innovative Applications of Artificial Intelligence, IAAI 2021, The Eleventh Symposium on Educational Advances in Artificial Intelligence, EAAI 2021, Virtual Event, February 2-9, 2021*, pages 8527–8537. AAAI Press, 2021. URL <https://ojs.aaai.org/index.php/AAAI/article/view/17035>.
- Nick Littlestone and Manfred K. Warmuth. The weighted majority algorithm. In *30th Annual Symposium on Foundations of Computer Science, Research Triangle Park, North Carolina, USA, 30 October - 1 November 1989*, pages 256–261. IEEE Computer Society, 1989. doi: 10.1109/SFCS.1989.63487. URL <https://doi.org/10.1109/SFCS.1989.63487>.
- Nick Littlestone and Manfred K. Warmuth. The weighted majority algorithm. *Inf. Comput.*, 108(2):212–261, 1994. doi: 10.1006/inco.1994.1009. URL <https://doi.org/10.1006/inco.1994.1009>.
- Kristian Lum and William Isaac. To predict and serve? *Significance*, 13(5):14–19, 2016.
- Celestine Mendler-Dünner, Juan C. Perdomo, Tijana Zrnic, and Moritz Hardt. Stochastic optimization for performative prediction. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/33e75ff09dd601bbe69f351039152189-Abstract.html>.
- John Miller, Juan C. Perdomo, and Tijana Zrnic. Outside the echo chamber: Optimizing the performative risk. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, volume 139 of *Proceedings of Machine Learning Research*, pages 7710–7720. PMLR, 2021. URL <http://proceedings.mlr.press/v139/miller21a.html>.
- Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dünner, and Moritz Hardt. Performative prediction. In *Proceedings of the 37th International Conference on Machine Learning, ICML 2020, 13-18 July 2020, Virtual Event*, volume 119 of *Proceedings of Machine Learning Research*, pages 7599–7609. PMLR, 2020. URL <http://proceedings.mlr.press/v119/perdomo20a.html>.

- Gregory Pierce and Donald Shoup. Sfpark: Pricing parking by demand. In *Parking and the City*, pages 344–353. Routledge, 2018.
- Mitas Ray, Lillian J. Ratliff, Dmitriy Drusvyatskiy, and Maryam Fazel. Decision-dependent risk minimization in geometrically decaying dynamic environments. In *Thirty-Sixth AAAI Conference on Artificial Intelligence, AAAI 2022, Thirty-Fourth Conference on Innovative Applications of Artificial Intelligence, IAAI 2022, The Twelveth Symposium on Educational Advances in Artificial Intelligence, EAAI 2022 Virtual Event, February 22 - March 1, 2022*, pages 8081–8088. AAAI Press, 2022. URL <https://ojs.aaai.org/index.php/AAAI/article/view/20780>.
- Shai Shalev-Shwartz. Online learning and online convex optimization. *Found. Trends Mach. Learn.*, 4(2):107–194, 2012. doi: 10.1561/22000000018. URL <https://doi.org/10.1561/22000000018>.
- Shai Shalev-Shwartz and Yoram Singer. Online learning meets optimization in the dual. In Gábor Lugosi and Hans Ulrich Simon, editors, *Learning Theory, 19th Annual Conference on Learning Theory, COLT 2006, Pittsburgh, PA, USA, June 22-25, 2006, Proceedings*, volume 4005 of *Lecture Notes in Computer Science*, pages 423–437. Springer, 2006. doi: 10.1007/11776420_32. URL https://doi.org/10.1007/11776420_32.
- Guanya Shi, Yiheng Lin, Soon-Jo Chung, Yisong Yue, and Adam Wierman. On-line optimization with memory and competitive control. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/ed46558a56a4a26b96a68738a0d28273-Abstract.html>.
- Arun Sai Suggala and Praneeth Netrapalli. Online non-convex learning: Following the perturbed leader is optimal. In Aryeh Kontorovich and Gergely Neu, editors, *Algorithmic Learning Theory, ALT 2020, 8-11 February 2020, San Diego, CA, USA*, volume 117 of *Proceedings of Machine Learning Research*, pages 845–861. PMLR, 2020. URL <http://proceedings.mlr.press/v117/suggala20a.html>.
- Richard S Sutton and Andrew G Barto. *Reinforcement learning: An introduction*. MIT press, 2018.
- Yuh-Shyang Wang, Nikolai Matni, and John C. Doyle. A system-level approach to controller synthesis. *IEEE Trans. Autom. Control.*, 64(10):4079–4093, 2019. doi: 10.1109/TAC.2018.2890753. URL <https://doi.org/10.1109/TAC.2018.2890753>.
- Dante Youla, Hamid Jabr, and Jr Bongiorno. Modern wiener-hopf design of optimal controllers—part ii: The multivariable case. *IEEE Transactions on Automatic Control*, 21(3):319–338, 1976.
- Lijun Zhang, Tianbao Yang, Rong Jin, and Zhi-Hua Zhou. Online bandit learning for a special class of non-convex losses. In Blai Bonet and Sven Koenig, editors, *Proceedings*

of the Twenty-Ninth AAAI Conference on Artificial Intelligence, January 25-30, 2015, Austin, Texas, USA, pages 3158–3164. AAAI Press, 2015. URL <http://www.aaai.org/ocs/index.php/AAAI/AAAI15/paper/view/9586>.

Peng Zhao, Guanghui Wang, Lijun Zhang, and Zhi-Hua Zhou. Bandit convex optimization in non-stationary environments. *J. Mach. Learn. Res.*, 22:125:1–125:45, 2021. URL <http://jmlr.org/papers/v22/20-763.html>.

Peng Zhao, Yu-Xiang Wang, and Zhi-Hua Zhou. Non-stationary online learning with memory and non-stochastic control. In Gustau Camps-Valls, Francisco J. R. Ruiz, and Isabel Valera, editors, *International Conference on Artificial Intelligence and Statistics, AISTATS 2022, 28-30 March 2022, Virtual Event*, volume 151 of *Proceedings of Machine Learning Research*, pages 2101–2133. PMLR, 2022. URL <https://proceedings.mlr.press/v151/zhao22a.html>.

A Proofs for Section 3

A.1 Proof of Lemma 3.1

We use proof by induction on T . The base case is $T = 0$. By definition, $x_1 = \arg \min_{x \in \mathcal{X}} R(x)$. Therefore, $R(x) \geq R(x_1)$ for all $x \in \mathcal{X}$. Using our notation defined earlier, we can write $g_0(x) \geq g_0(x_1)$ for all $x \in \mathcal{X}$. Now, assume that the lemma is true for $T - 1$. That is,

$$\sum_{t=0}^{T-1} g_t(x) \geq \sum_{t=0}^{T-1} g_t(x_{t+1}). \quad (18)$$

Let $x \in \mathcal{X}$ be arbitrary. Since $x_{T+1} = \arg \min_{x \in \mathcal{X}} \sum_{t=0}^T g_t(x)$, we have

$$\sum_{t=0}^T g_t(x) \geq \sum_{t=0}^T g_t(x_{T+1}) \quad (19)$$

$$= \sum_{t=0}^{T-1} g_t(x_{T+1}) + g_T(x_{T+1}) \quad (20)$$

$$\geq \sum_{t=0}^{T-1} g_t(x_{t+1}) + g_T(x_{T+1}) \quad \text{by Eq. (18)} \quad (21)$$

$$= \sum_{t=0}^T g_t(x_{t+1}). \quad (22)$$

This shows that the lemma is true for T and completes the proof.

A.2 Proof of Lemma 3.2

Note that

$$\sum_{t=1}^T g_t(x_t) - \sum_{t=1}^T g_t(x^*) \leq \sum_{t=1}^T g_t(x_t) - \sum_{t=1}^T g_t(x^*) + g_0(x^*) - g_0(x_1) \quad (23)$$

because $x_1 = \arg \min_{x \in \mathcal{X}} g_0(x)$. The proof of this lemma now follows by using the above inequality, $x = x^*$ in Lemma 3.1, the definition $g_0 = \frac{R}{\eta}$, and the definition of D .

A.3 Proof of Lemma 3.3

Using Lemma 3.2, we have

$$\begin{aligned} & \sum_{t=1}^T g_t(x_t) - \sum_{t=1}^T g_t(x^*) \\ & \leq \sum_{t=1}^T \underbrace{g_t(x_t) - g_t(x_{t+1})}_{(a)} + \frac{D}{\eta}. \end{aligned} \quad (24)$$

We can bound the term (a) as follows. Define $F_t(x) = \sum_{s=0}^{t-1} g_s(x)$. Then, $x_t = \arg \min_{x \in \mathcal{X}} F_t(x)$ and $x_{t+1} = \arg \min_{x \in \mathcal{X}} F_{t+1}(x)$. Since $\{g_s\}_{s=1}^T$ are convex, R is α -strongly-convex, and $g_0 = \frac{R}{\eta}$, this implies F_t is $\frac{\alpha}{\eta}$ -strongly-convex. So,

$$\begin{aligned} F_t(x_{t+1}) & \geq F_t(x_t) + \frac{\alpha}{2\eta} \|x_{t+1} - x_t\|^2, \\ F_{t+1}(x_t) & \geq F_{t+1}(x_{t+1}) + \frac{\alpha}{2\eta} \|x_{t+1} - x_t\|^2. \end{aligned}$$

Adding the above two equations yields

$$g_t(x_t) - g_t(x_{t+1}) \geq \frac{\alpha}{\eta} \|x_{t+1} - x_t\|^2. \quad (25)$$

By our assumptions about the convexity (Assumption A3) and norms on the gradients (Assumption A2) of g_t , we also have

$$\begin{aligned} g_t(x_t) - g_t(x_{t+1}) & \leq \langle \nabla g_t(x_t), (x_t - x_{t+1}) \rangle \\ & = \langle \nabla \check{f}_t(x_t), (x_t - x_{t+1}) \rangle \\ & \leq \|\nabla \check{f}_t(x_t)\| \|x_t - x_{t+1}\| \\ & \leq L \|x_{t+1} - x_t\|. \end{aligned} \quad (26)$$

Combining Eqs. (25) and (26), we have

$$\|x_{t+1} - x_t\| \leq \eta \frac{L}{\alpha}. \quad (27)$$

Using this in Eq. (26), we have

$$g_t(x_t) - g_t(x_{t+1}) \leq \eta \frac{L^2}{\alpha}. \quad (28)$$

Finally, substituting Eq. (28) into Eq. (24) yields the desired result.